

JPAM POLICY INSIGHTS

Failing to Learn From Failure in Online Credit Recovery Assessments

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ABSTRACT

Technology-facilitated courses, completed online for the purpose of credit recovery (OCR), are the most common means through which US high school students retake courses required for high school graduation. Yet a growing body of research has raised concerns regarding student learning in these courses, with low-quality assessments posited as a limiting factor. As course assessments are fundamental to evaluating student mastery and supporting student learning, we investigate this concern by systematically reviewing every assessment item in the most widely used OCR course, Algebra 1. We also examine pathways for passing the course mastery tests that allow students to bypass learning the content. In addition, we identify policy options and practices that state and local agencies are implementing or could adopt to regulate and improve OCR. We find that OCR assessments as executed lack rigor and validity, and that answers to most assessment items are readily available across multiple websites or platforms. We offer recommendations to improve rigor, close pathways that call into question the validity of results, strengthen implementation procedures, and increase state-level oversight of OCR providers.

1 | Introduction

Credit recovery courses provide students with the opportunity to recover credit required for high school graduation by repeating a previously failed course. Before the pandemic, 71% of credit recovery courses nationally were delivered online (US Department of Education 2018). High rates of course failure and chronic absenteeism have contributed to a recent surge in online credit recovery (OCR) adoption in schools that is expected to continue well into the future (Barshay 2022; Reeves 2022). Moreover, some studies have identified OCR as a major contributor to rising graduation rates (Heinrich and Darling-Aduana 2021; Powell et al. 2015; Tyner and Munyan-Penney 2018), suggesting that OCR may be viewed as an attractive “quick

fix” to the problem of more students at risk of dropping out since the pandemic (Shen-Berro 2023).

Although some empirical analyses confirm positive associations between OCR, credits earned, and high school graduation rates, the research also finds insignificant or negative relationships between OCR and student achievement, postsecondary graduation, and post-high school earnings (Heinrich et al. 2019; Heinrich and Darling-Aduana 2021; Heinrich and Cheng 2022; Heppen et al. 2017; Rickles et al. 2018; Rickles et al. 2023; Viano and Henry 2024). More generally, policies that provide academic leniency as a means to increase high school graduation rates, as documented in the implementation of OCR, are associated with decreased student effort, attendance, and test score gains (Bowden et al.

2023; Coffey and Tyner 2024; Goldhaber and Goodman Young 2024). Research on OCR implementation documents a lack of meaningful student engagement in instruction and assessments (including online searching for test answers), and a lack of supervision and instructional supports for students (Heinrich et al. 2019; Darling-Aduana et al. 2019; Heinrich and Darling-Aduana 2021; Hess 2019).

While this problem extends beyond credit recovery courses, the issue is more urgent in a system that encourages students to recover high school credits for graduation as “quickly and cheaply as possible—regardless of what they have learned” (Horn 2017, 2). Assessments are critical to OCR courses, as frequent assessments are the primary means through which course progression is achieved. Educational assessment also informs formative feedback and targeted support to improve the learning experience (Ashford-Rowe et al. 2014; Stassen et al. 2001). For these reasons, the validity and rigor of assessments have important implications for whether students are truly learning the skills they need to be successful later in life.

We accordingly explored the rigor, quality, and implications of OCR assessments with a focus on one of the most commonly taken courses, Algebra I, developed by one of the largest online course vendors in the United States, Edmentum. Courses sold by this vendor are used by over 5 million students across all 50 states. We chose to focus on Algebra I due to the importance of the course as a gatekeeper to more advanced mathematics coursework and its strong relationship with subsequent academic and career outcomes (Heppen et al. 2017). Through this exploration, we identified several critical problems with OCR assessments and their implementation that question the usefulness of these assessments as a gauge for student learning.

- OCR tests require minimal engagement in higher-order tasks.
- Test answers are readily accessible online.
- Pretest performance is not a good predictor of mastery.
- Retakes are common and make extensive use of repeat questions.
- Few states provide adequate OCR policy or regulation.

Although we focus our analysis specifically on OCR, many of these assessment strategies are observed across online instructional tools and platforms (Darling-Aduana et al. 2025; Boninger et al. 2017; Daniels et al. 2021). In documenting these practices, we aim to extend understanding of OCR assessments from the item level to the policy level and illuminate micro, meso, and macro strategies for increasing the rigor and validity of OCR assessments, with the ultimate goal of supporting student learning in these courses. We conclude with a discussion of concrete policy actions that could be taken to make online course assessments more meaningful and increase student opportunities to learn in OCR courses and assessments.

2 | OCR Assessment Quality and Rigor

2.1 | OCR Tests Require Minimal Engagement in Higher-Order Tasks

Our descriptive analysis of 1408 assessment items in the OCR Algebra I course, using Krathwohl (2002)’s update of Bloom’s Taxonomy for classifying educational learning objectives by their levels of complexity and specificity (Bloom et al. 1956), highlights the limited nature of the OCR assessments (see Figure 1). All question items required students to recognize or recall content, and about three-fourths of the questions asked students to apply content (e.g., solve or simplify an equation). However, only 5% of the OCR Algebra I questions required students to analyze content and just 4% of the assessment questions required students to create content. None of the assessment items involved evaluating content. In total, less than 10% of the questions required students to engage in any higher-order tasks, that is, to either analyze, evaluate, or create content in answering the question. Although rare, questions requiring students to apply their knowledge—for instance, of equations’ characteristics or to draw a graph or create or solve equations based on real-life scenarios—were most likely to foster these higher-order tasks.

Potentially contributing to the prevalence of lower-order tasks, 83% of assessment items are multiple-choice questions (primarily single-response questions). This does not vary significantly by assessment type. The mastery tests that follow each lesson consist of five questions, providing a narrow (and insufficient in most cases) assessment of the wide range of knowledge and skills covered in a given lesson.

2.2 | Test Answers Are Readily Accessible Online

Moreover, the answers to the questions are readily available on multiple websites or platforms. In fact, 91% of the answers to the questions could be found in a Google search, and 82% of the answers found were correct (did not require further searching to get the correct answer). The answers to the questions were encountered on 20 different websites, although Brainly was the first hit in 95% of searches. Some answers had been published online as early as 2015, with more than 70% posted in 2020 or earlier. This implies that most answers to the OCR Algebra I questions were available as schools shifted more students to online learning during the pandemic. This also suggests that the online course vendor was not regularly updating assessment items.

Next, we examined the extent to which students could generate correct answers using generative AI (as of 2023). When the assessment items were pasted verbatim into ChatGPT, correct answers were generated in about one third of the responses. Some additional interaction with the tool by modifying prompts was required to get the correct answer for another 22% of question items. Since conducting this study, ChatGPT has made improvements to how the generative AI platform reads and solves equations, which may result in more correct answers. However, regardless of ChatGPT effectiveness, a simple Internet search appears sufficient to find answers to most questions.

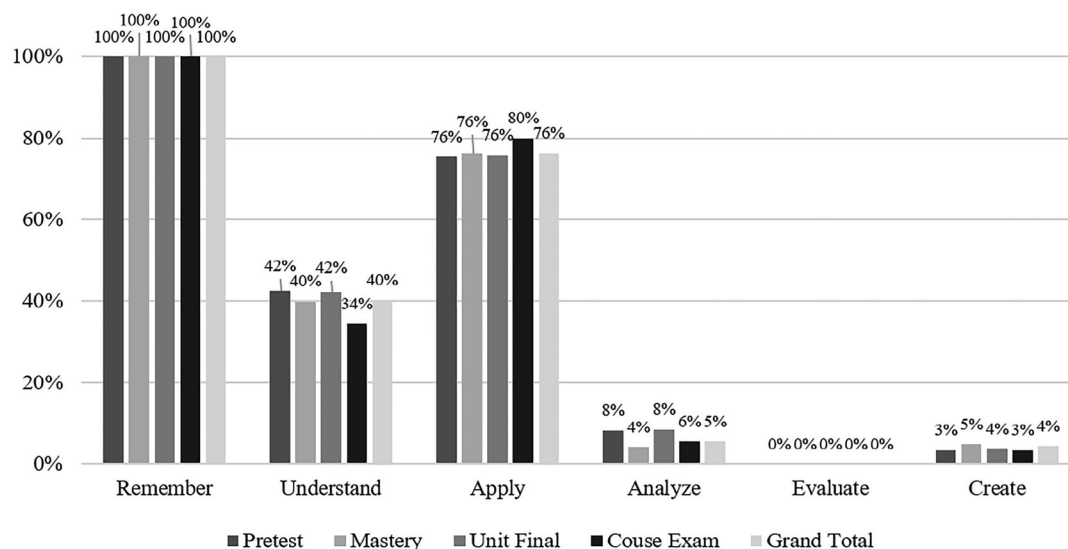


FIGURE 1 | Percentage of items that engaged each level of Bloom's taxonomy by test type.

2.3 | Pretest Performance Is Not a Good Predictor of Mastery

In addition, as indicated earlier, students take pretests that may allow them to move faster through the online course by testing out of lessons. However, the students will still be tested on the lesson material in a summative posttest. Analysis of the sampled assessment data showed that pretest performance was not a good predictor of performance on the course material in a summative posttest. In only 38% percent of cases did a "mastery" level score (80 or higher) on a pretest correspond to a passing score (75 or higher) for a given topic on the summative posttest, raising concerns about the extent to which pretest outcomes reflect students' mastery of a topic.

3 | OCR Practice and Policy

3.1 | Retakes Are Common and Make Extensive Use of Repeat Questions

Even though the assessments generally involved lower-order tasks, analysis of summary statistics within the online course system showed that the majority of on-campus OCR students failed the mastery tests on the first attempt (modal score = 40), requiring a retake. Analysis of 200 randomly sampled assessments revealed that on test retakes, two to three of the five question items were typically the same questions as in the first assessment (for nearly two thirds of the assessments analyzed). In fact, four of five questions were repeated on retakes in more than 8% of the 200 mastery tests. In less than 6% of assessments, no questions were repeated, and in 22% of assessments, just one question was repeated on the retake. Posttests and final exams were repeated entirely on retakes.

The OCR mastery tests are created from test banks large enough for two or four entirely new tests. Yet by design, the online course-taking system generates repeat questions, which makes it easier for students to pass the test on the second attempt. As imple-

mented in our study context, the system allowed students up to three retakes on a mastery test (that was unlocked automatically upon completion of the lesson), so that students working on them at home could take the tests without supervision. For posttest retakes, the only option available in the online course-taking system (as implemented) was for the instructor to reset the test, which erased the original score and allowed the student to start again with the same questions after the student had viewed the right and wrong answers on the initial test.

3.2 | Few States Provide Adequate OCR Policy or Regulation

Our descriptive findings on OCR course assessments and student learning call for policies and regulations to support more accurate, fair, and meaningful OCR assessments. While about 59% of states had some policy or regulation for online learning and 27% addressed credit recovery as of Fall 2023, less than 30% had any policy specific to OCR; 35% of states had no policies on online learning, credit recovery or OCR (see Figure 2).

Among the 30 states with policies or directives that explicitly mentioned online learning, policy content was varied. Some included provisions addressing providers and their accreditation (about 50%) or pertained to curricula (about 40%), with over a quarter of these states addressing both providers and curricula. Some policies circumscribed virtual schooling, distance learning, and remote learning, and some were directed toward creating new online schooling opportunities, including virtual schools and networks. In addition, some policies established state boards to oversee district implementation of online learning and to accredit outside providers.

Of the 14 states that had established guidelines for credit recovery, credit recovery was often described as a form of alternative learning that could assist students with differential needs and abilities to complete their coursework. About half of these policies included a provision that explicitly required alignment between

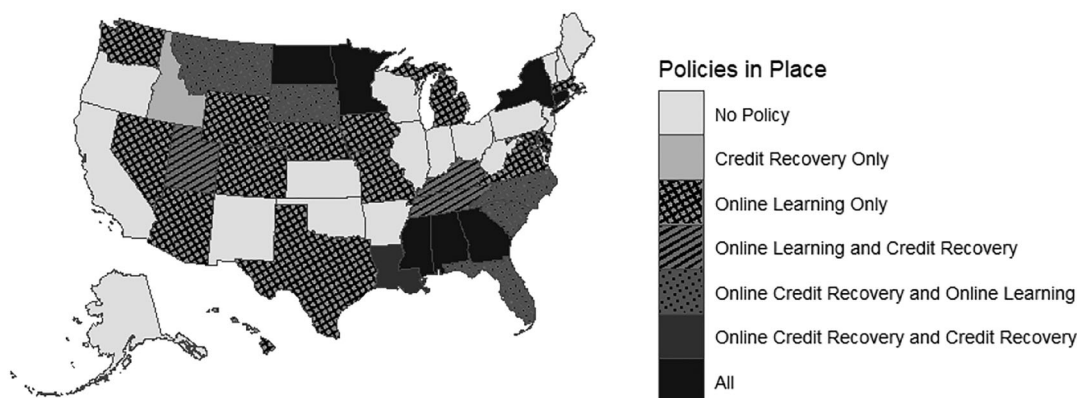


FIGURE 2 | Online and credit recovery related policies by state (as of fall 2023).

the credit recovery curriculum and the state curriculum; however, few states specified provisions to ensure that credit recovery providers were properly accredited. Among 15 states with policies explicitly addressing OCR, there was only one state that included provisions for both the type of curriculum and the type of provider offering OCR; New York had policies with specific language regarding the quality of OCR courses, including that they should involve “regular and substantive” interactions between students and teachers.

Since 2023, only New York has taken concerted actions to regulate OCR vendors. New York state OCR teachers of record are now required to be employees of the public school district, while New York City schools can only use preapproved virtual content vendors and subject-certified teachers (not vendor-provided instructors). In Texas, the Texas Education Agency is currently engaged in a rulemaking process that will establish vendor standards for virtual education, following the 2025 passage of Senate Bill 569.

4 | Summary

A central objective of educational assessment is to improve student learning by evaluating students’ mastery of educational content and identifying students who may require additional support to meet course aims. One of the expressed concerns about OCR is that it is focused on helping students recover course credits as quickly as possible—using pretesting to bypass lessons, allowing students to drive the instruction and access to assessments in the classroom and/or at home, utilizing brief assessments, and allowing numerous test retakes—regardless of whether learning is taking place. Our study investigated the merits of this concern using the assessments in a widely utilized course (Algebra I) offered through an OCR program used in all 50 states to understand their nature (details) and validity, as well as how they are implemented and those implications for their validity.

Our study findings show that most of the assessments included in the Algebra I course (greater than 80%) were multiple-choice, single-response questions, with high re-use of the same questions on the frequent test retakes. Moreover, the test questions were limited in their assessment of students’ higher cognitive skills;

less than 10% of the questions required students to analyze, evaluate, or create content. In addition, the students could avoid applying *any* skills in completing the assessments by taking advantage of the wide availability of test answers on multiple online platforms. More than 90% of the answers to test questions could be found using Google, and we encountered 20 different websites or platforms where the answers could be located. A large majority of the answers had been posted online for many years. The standard OCR system settings (frequent repetition of questions, multiple test retakes), lack of state or district policy or regulation safeguards, and minimal monitoring of student interactions with the course (and their use of external websites or platforms) in the typical online learning implementation model further diminish the value or utility of the OCR course assessments.

Although this research is limited in its study of a particular mathematics course as implemented in a single school district, prior research has documented that its implementation as observed in this setting is typical of how OCR courses are enacted by the largest online course providers (Heinrich et al. 2019; Heppen et al. 2017; Rickles et al. 2018, 2023; Viano 2021; Viano and Henry 2024). Generalizability is also strengthened by the examination of a course developed by one of the largest online course providers in the country, implying that changes made to improve assessments in the OCR course system could affect students taking the course nationwide. This makes the lack of explicit policies and regulations on the accreditation of OCR providers of considerable concern. More broadly, our findings support growing evidence that educational policies and practices promoting academic leniency as a means to boost graduation rates may be harming the learning experiences and outcomes of the very students those policies are designed to help (Bowden et al. 2023; Coffey and Tyner 2024; Goldhaber and Goodman Young 2024).

5 | Implications for Policy and Practice

Our findings suggest several areas for improvement in the design and implementation of OCR assessments, many of which may apply to online assessments more broadly, particularly those developed by large, for-profit online course vendors. First, assessments should include items that test understanding at a range of

cognitive levels, including items that require students to analyze, evaluate, and create. This may require the inclusion of more open-ended and project-based assignments by vendors or as supplemental activities designed and implemented by school-based educators (Darling-Aduana 2021). Although it could be argued that some state-level and other standardized assessments also fail to assess students on higher cognitive tasks, reducing assessment to the lowest common denominator is hardly a compelling reason for not requiring more from companies profiting from public education dollars. In addition, there are other testing companies that have found techniques to assess cognitively demanding tasks (i.e., GRE) that could be expanded to OCR and other standardized assessments (at grade-appropriate levels).

Second, efforts should be made to eliminate opportunities to answer items correctly without understanding the underlying content. District testing procedures worth considering include in-person proctoring, honor code reminders, reducing testing policy loopholes (i.e., opportunities to retake the exact same exam after viewing answers after the initial attempt), and limiting access to outside resources during testing (Dendir and Maxwell 2020; Mukherjee et al. 2023; Noorbehbahani et al. 2022).

In terms of their implications for policy, the limitations identified of OCR assessment in evaluating true learning suggest that additional rigor and regulation regarding OCR provider accreditation and course alignment with state curriculum standards appear merited (Glass and Welner 2011; Kirsch and Smiley 2017; Watson and Gemin 2008). At a minimum, states should strive to have policies or regulations in place that ensure minimum standards for both online learning and credit recovery, if not OCR-specific policies. Items to address include mandating the regular updating of course materials, including assessment items, and instituting procedures to ensure the use of varying test versions, similar to the practices large-scale assessment companies use to minimize cheating.

Another option to consider that also applies to credit recovery courses offered in traditional classroom settings is to require students to take specific, statewide end-of-course exams in order to graduate, as six states (Georgia, Illinois, Maryland, Mississippi, Missouri, and Tennessee) currently do (FairTest 2024). For example, if Algebra I was a required end-of-course exam in states using the OCR Algebra I course, school districts might be more motivated to institute policies and checks to ensure students are learning in (and not just passing) OCR courses.

In the meantime, school districts can fill policy gaps by developing their own processes for evaluating the rigor and usefulness of OCR courses, including OCR assessments (Tyner and Munyan-Penney 2018; Malkus 2019), which may include the use of existing tools such as the Quality Matters Online Course Design Rubric to evaluate online courses (see also Darling-Aduana 2021) and incorporating online course adoption into existing curriculum review processes. Taken together, the implementation of the proposed policies and practices can support the development and administration of rigorous, valid OCR assessments that achieve their aim of providing valuable tools to accurately evaluate and support student learning.

Conflicts of Interest

The authors declare no conflicts of interest.

Ethics Statement

This study underwent university IRB review (#230858), and it was determined that this was not human subjects research.

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The authors have nothing to report.

Data Availability Statement

The datasets generated and analyzed in this study are available in Excel format from the corresponding author upon request and from the Harvard Dataverse Network at <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/LX610H>.

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